

Systems of First Order Linear ODEs

1 First order ODE systems

The general system of first order equations involving n dependent variables defined on an interval $I \subseteq \mathbb{R}$ can be written as

$$\begin{cases} y_1'(t) = F_1(t, y_1, \dots, y_n) \\ y_2'(t) = F_2(t, y_1, \dots, y_n) \\ \vdots \\ y_n'(t) = F_n(t, y_1, \dots, y_n) \end{cases} \quad \text{for } t \in I. \quad (1)$$

with initial conditions $y_1(t_0) = x_1, \quad y_2(t_0) = x_2, \quad \dots, \quad y_n(t_0) = x_n,$

The solution to the system can be expressed in vector form as

$$\mathbf{y}(t) = \begin{pmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_n(t) \end{pmatrix}$$

Theorem 1.1 (Existence and Uniqueness).

For the system of first order ODEs (1), if the functions $F_1, F_2, \dots, F_n$ and the partial derivatives $\frac{\partial F_i}{\partial y_j}$ for all pairs $i, j = 1, 2, \dots, n$ are continuous on a region R defined by

$$R := [t_0 - a, t_0 + a] \times [x_1 - b, x_1 + b] \times \dots \times [x_n - b, x_n + b] \subseteq \mathbb{R}^{n+1},$$

where constants $a, b > 0$, then there exists a unique solution $\mathbf{y}(t)$ for $t \in (t_0 - h, t_0 + h)$ to the IVP, where

$$h = \min\{a, b/M\} \quad \text{and} \quad M = \max_{\mathbf{v} \in R} \{|F_1(\mathbf{v})|, |F_2(\mathbf{v})|, \dots, |F_n(\mathbf{v})|\}.$$

Any n -th order ODE can be rewritten as a system of n first order ODEs. Given an n -th order ODE

$$y^{(n)} = F(t, y, y', \dots, y^{(n-1)}),$$

setting $z_1 = y, z_2 = y', \dots, z_n = y^{(n-1)}$, we obtain the system

$$\begin{cases} z_1' = z_2 \\ z_2' = z_3 \\ \vdots \\ z_{n-1}' = z_n \\ z_n' = F(t, z_1, z_2, \dots, z_n) \end{cases}$$

which is a system of n first order ODEs.

Definition 1.1 (First-Order Linear System).

A system of first order ODEs is called **linear** if it can be expressed in the form

$$\mathbf{y}'(t) = \mathbf{P}(t)\mathbf{y}(t) + \mathbf{g}(t) \quad \text{for } t \in I, \quad (2)$$

where

$$\mathbf{y}(t) = \begin{pmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_n(t) \end{pmatrix} \quad \mathbf{g}(t) = \begin{pmatrix} g_1(t) \\ g_2(t) \\ \vdots \\ g_n(t) \end{pmatrix} \quad \mathbf{P}(t) = \begin{pmatrix} p_{11}(t) & p_{12}(t) & \dots & p_{1n}(t) \\ p_{21}(t) & p_{22}(t) & \dots & p_{2n}(t) \\ \vdots & \vdots & \ddots & \vdots \\ p_{n1}(t) & p_{n2}(t) & \dots & p_{nn}(t) \end{pmatrix}$$

Definition 1.2 (Homogeneous First-Order Linear System).

A first order linear system is called **homogeneous** if $\mathbf{g}(t) = \mathbf{0}_n$ for all $t \in I$, i.e., it can be expressed as

$$\mathbf{y}'(t) = \mathbf{P}(t)\mathbf{y}(t) \quad \text{for } t \in I.$$

Theorem 1.2 (Existence and Uniqueness for First-Order Linear Systems).

For the first-order linear system (2), if all entries of $\mathbf{P}(t)$ and $\mathbf{g}(t)$ are continuous on an interval I , then for any $t_0 \in I$ and $\mathbf{y}_0 \in \mathbb{R}^n$, there exists a unique solution $\mathbf{y}(t)$ to the IVP

$$\begin{cases} \mathbf{y}'(t) = \mathbf{P}(t)\mathbf{y}(t) + \mathbf{g}(t) \\ \mathbf{y}(t_0) = \mathbf{a} \in \mathbb{R}^n \end{cases} \quad \text{for } t \in I.$$

In the rest parts, we will focus on the following homogeneous system defined on an open interval $I \subseteq \mathbb{R}$:

$$\mathbf{y}'(t) = \mathbf{P}(t)\mathbf{y}(t) \quad \text{for } t \in I, \quad (3)$$

where $\mathbf{y} \in C^1(I)^n$ and $\mathbf{P} \in C(I)^{n \times n}$, i.e., all entries of $\mathbf{P}$ and $\mathbf{y}$ are continuous on I .

Theorem 1.3 (Principle of Superposition).

If $\mathbf{y}_1(t)$ and $\mathbf{y}_2(t)$ are solutions to the homogeneous first-order linear system (3), then the linear combination

$$\mathbf{y}(t) = c_1\mathbf{y}_1(t) + c_2\mathbf{y}_2(t)$$

is also a solution for any constants c_1, c_2 .

Definition 1.3 (Wronskian).

The **Wronskian** of n vector functions $\mathbf{f}_1(t), \mathbf{f}_2(t), \dots, \mathbf{f}_n(t) \in C(I)^n$ is defined as

$$W(\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_n)(t) := \det \left(\begin{bmatrix} \mathbf{f}_1(t) & \cdots & \mathbf{f}_n(t) \end{bmatrix} \right) \quad \text{for } t \in I.$$

Remark. The Wronskian for vector functions is different from the Wronskian for scalar functions. There is no derivative involved in the definition of the Wronskian for vector functions.

Theorem 1.4 (Liouville's Formula).

Let $\mathbf{y}^{(1)}(t), \mathbf{y}^{(2)}(t), \dots, \mathbf{y}^{(n)}(t)$ be n solutions to the homogeneous system (3). Then the Wronskian of these solutions is given by

$$W(\mathbf{y}^{(1)}, \mathbf{y}^{(2)}, \dots, \mathbf{y}^{(n)})(t) = c \cdot \exp \left(\int \text{tr}(\mathbf{P}(t)) dt \right) \quad \text{for } t \in I,$$

where c is a constant not depending on t .

Remark. The **trace** of a square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is defined as the sum of its diagonal entries:

$$\text{tr}(\mathbf{A}) := a_{11} + a_{22} + \cdots + a_{nn}.$$

Proof (Liouville's Formula).

Before the proof, we need to introduce several theorems from linear algebra:

1. For any square matrix $\mathbf{A}$,

$$\frac{d}{dt}(\det(\mathbf{A})) = \text{tr} \left(\text{adj}(\mathbf{A}) \frac{d\mathbf{A}}{dt} \right). \quad (4)$$

2. The trace remains unchanged under cycle permutations:

$$\text{tr}(\mathbf{ABC}) = \text{tr}(\mathbf{BCA}). \quad (5)$$

3. Property of the adjugate matrix:

$$\text{adj}(\mathbf{A})\mathbf{A} = \mathbf{A} \text{adj}(\mathbf{A}) = \det(\mathbf{A})\mathbf{I}. \quad (6)$$

Now, let $\mathbf{Y} := [\mathbf{y}^{(1)} \quad \mathbf{y}^{(2)} \quad \cdots \quad \mathbf{y}^{(n)}] \in C^1(I)^{n \times n}$ and $W := \det(\mathbf{Y})$ be the Wronskian. Then we have

$$\begin{aligned} W' &= \text{tr}(\text{adj}(\mathbf{Y})\mathbf{Y}') \\ &= \text{tr}(\text{adj}(\mathbf{Y})\mathbf{PY}) \quad \text{since } \mathbf{y}' = \mathbf{Py} \\ &= \text{tr}(\mathbf{Y} \text{adj}(\mathbf{Y})\mathbf{P}) \\ &= \text{tr}(\mathbf{WIP}) \\ &= W \text{tr}(\mathbf{P}). \end{aligned}$$

Hence the Liouville's formula. □

Theorem 1.5.

If $\mathbf{y}^{(1)}, \mathbf{y}^{(2)}, \dots, \mathbf{y}^{(n)}$ are n solutions to a homogeneous system (3), then

$$\forall t \in I, \quad W(\mathbf{y}^{(1)}, \mathbf{y}^{(2)}, \dots, \mathbf{y}^{(n)})(t) = 0 \quad \Longleftrightarrow \quad \exists t_0 \in I \text{ s.t. } W(\mathbf{y}^{(1)}, \mathbf{y}^{(2)}, \dots, \mathbf{y}^{(n)})(t_0) = 0.$$

Proof.

Proven by Liouville's formula. □

Theorem 1.6.

If the n solutions $\mathbf{y}^{(1)}(t), \mathbf{y}^{(2)}(t), \dots, \mathbf{y}^{(n)}(t)$ of the homogeneous system (3) are **linearly independent** for each $t \in I$, then every solution of the system can be expressed as a linear combination:

$$\mathbf{y}(t) = c_1 \mathbf{y}^{(1)}(t) + c_2 \mathbf{y}^{(2)}(t) + \dots + c_n \mathbf{y}^{(n)}(t) \quad \text{for } t \in I$$

in **exactly one way**.

Definition 1.4 (Fundamental Set of Solutions).

A set of n **linearly independent** solutions $\{\mathbf{y}^{(1)}(t), \mathbf{y}^{(2)}(t), \dots, \mathbf{y}^{(n)}(t)\}$ to the homogeneous system (3) is called a **fundamental set of solutions**.